DIGITAL TWIN CHALLENGE 2026 | HAPPIEST HEALTH

PulseForge

Privacy-preserving, explainable health-risk monitoring with federated learning



Team PulseForge | Pranveer Singh Institute of Technology, Kanpur

Health AI is stuck on trust



Centralised data is a liability

Pooling wearable and hospital records on one server creates a single point of failure. One breach exposes thousands of patients.



Data stays siloed

Hospitals and users rightly refuse to share raw records, so models are trained on small, biased samples.



Black-box risk scores

A label with no reasons is hard for doctors to trust or act on, and hard for patients to accept.

THE GAP

Privacy, accuracy and explainability are usually traded against each other.

PulseForge delivers all three.

Why health, and why now



Most sensitive data

Health records reveal conditions, habits and identity. Misuse cannot be undone.



Always-on sensing

Wearables stream vitals around the clock, enabling early warning before symptoms escalate.



High stakes

A wrong or unexplained alert can delay care or cause needless worry.



Strict regulation

GDPR, HIPAA and India's DPDP Act 2023 push data minimisation by design.

Health is where privacy-preserving AI matters most, and where it is hardest to get right.

Learn together. Share nothing raw.



Federated Learning

Each device trains on its own data. Only model weights travel; a server averages them into one shared model.



Differential Privacy

Calibrated Laplace noise is added to the weights before release (ε = 1.0), so no single record can be inferred.



Explainable AI

SHAP shows the top factors behind every prediction, computed on the device and never on a server.

Output: a risk class (Normal, Mild, Moderate or Severe) with its top-3 reasons, delivered on the user's device by a personalised model that acts as their private health twin.

Raw data never leaves the device

ON DEVICE

Collect

Wearable or Google Fit vitals: heart rate, SpO2, BP, glucose, sleep, stress.

ON DEVICE

Protect

Local model trains on-device. Laplace noise is added once, at the final round.

SERVER

Aggregate

Size-weighted FedAvg over noisy weights. Mean and variance only for scaling.

ON DEVICE

Personalise

Global model is fine-tuned locally into the user's private health twin.

ON DEVICE

Explain and alert

Risk class plus SHAP top-3 factors become an alert on the phone.

Private zone: raw data stays here

Sees noisy weights only

Private zone: inference on device

Weights flow device to server to device. Raw data and explanations never travel.

How we built it

6,000
records: 2,314 real + 3,686 SMOTE

17
features: 13 vitals + 4 engineered

4
risk classes: Normal to Severe

3
non-IID clients (Dirichlet alpha = 1.0)

Build pipeline

1. Split 6,000 records into 3 non-IID client partitions
2. Share only mean and variance so every client scales features identically
3. Train an L2-regularised logistic regression locally (SGD, 5 rounds)
4. Add Laplace noise once, on the final round (ε = 1.0)
5. Size-weighted FedAvg, then SHAP explanations on each device

Tech stack

- Python 3
- scikit-learn
- SHAP
- NumPy / pandas
- Matplotlib
- GitHub / Kaggle

Privacy by design

SINGLE-RELEASE LAPLACE MECHANISM

$$\Delta f = 2R / (N \cdot \alpha)$$

sensitivity of the local model

$$b = \Delta f / \varepsilon$$

noise scale

$$\tilde{w} = w + Lap(0, b)$$

weights released to the server

R clip = 1.5, α = 0.20, ε = 1.0 gives b ≈ 0.009 per client. Noise is added only in the last round, so FedAvg converges cleanly first.

What leaves the device?

Item	What is shared
Raw patient data	**Never**
Feature statistics	Mean and variance only
Model update	Noisy weights only
SHAP explanations	Stay on the device
Global importance	Post-processing: no extra privacy cost

The server only ever sees noisy weights, never a patient record.

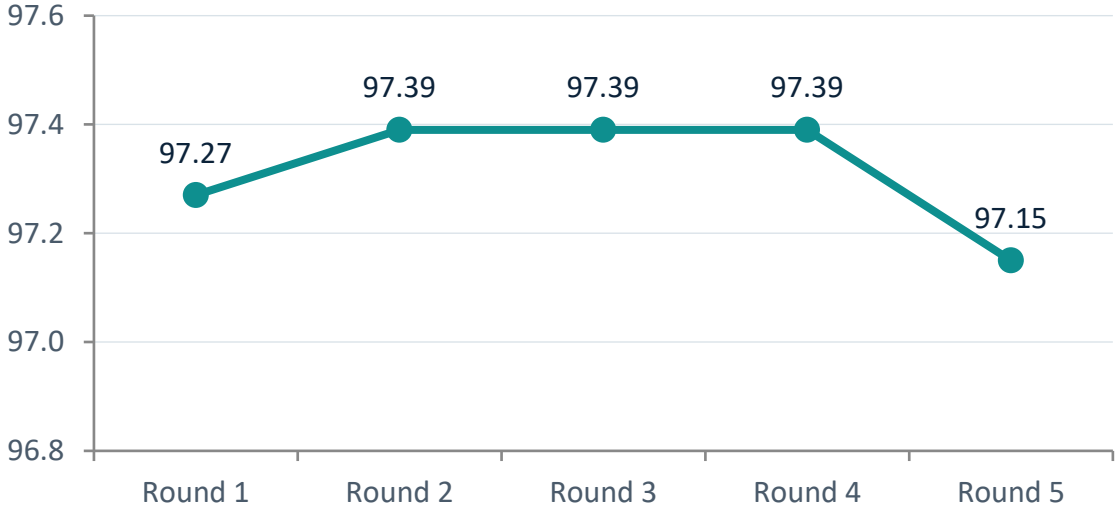
Privacy costs under a quarter point

97.15%
global accuracy with ε = 1.0 privacy

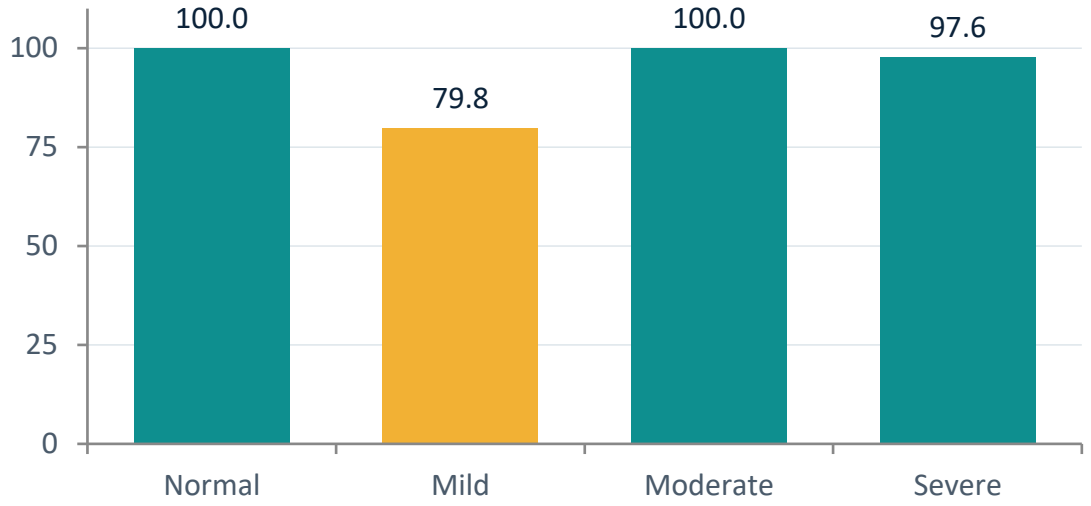
0.24 pt
accuracy cost of privacy (non-private: 97.39%)

0
raw patient records shared with the server

Global accuracy per federated round (%)



Recall per risk class (%)



Noise is applied only in round 5. Mild overlaps with Severe: 17 of 104 Mild cases.

Every prediction comes with reasons

Normal

Stress level 0.39
Sleep quality 0.37

Lifestyle and autonomic signals

Mild Event

HRV (SDNN) 0.44
Blood oxygen 0.27

Heart-rate variability and oxygenation

Moderate Event

Glucose level 0.55
Diastolic BP 0.37

Glucose-led metabolic pattern

Severe Event

Systolic BP 0.51
Diastolic BP 0.49

Blood-pressure crisis pattern

ON-DEVICE ALERT (SAMPLE)

Normal

Top factors behind this prediction

Stress level ↑
HRV (SDNN) ↑
HR-to-stress ratio ↑

↑ supports the predicted class. Computed on the phone with SHAP; global importance uses already-noised weights, so it costs no extra privacy.

Numbers: global feature importance (absolute weight) from the DP-protected model.

Backed by published research

PUBLISHED — IEEE MPCON 2026

Differential Privacy for Medical Data Analysis

Laplace DP, random-forest risk prediction and blockchain-logged alerts on 25,000 records (8 respiratory diseases).

G. Misra, B. Hazela, B. K. Chaurasia, A. Singh
DOI 10.1109/MPCON69668.2026.11508248

SUBMITTED — IEEE

Federated Learning with Differential Privacy and Explainable AI for Secure Healthcare Data Analysis

This work: FedAvg, single-release DP and on-device SHAP. 97.15% accuracy at $\varepsilon = 1.0$.

A. Singh, G. Bajpai, A. Bajpai, A. P. Tripathi

OPEN SOURCE — GitHub and Kaggle

PulseForge code and trained model

Code, trained model, plots and reproducible runs (fixed seed 42).

github.com/Anurag-kushwaha2207/PulseForge_Pranveer-Singh-Institute-of-Technology
kaggle.com/m/744851

Our published work applied differential privacy in a centralised pipeline. PulseForge extends it to many devices, with explanations.

From simulation to real users

DONE

Validated on 3 clients

FL, DP and XAI pipeline reaches 97.15% accuracy. Code released; one paper published, one submitted.

NEXT

Real wearable data

Google Fit consent flow: each user's account becomes one federated client.

THEN

Live service

Hosted FedAvg server, client sampling for many users, clinician dashboard.

LATER

Clinical validation

Multi-condition risk labels and clinician-reviewed explanations.

Research prototype, not a medical device: clinical validation comes before any deployment.

Team PulseForge

Team Leader

Anurag Singh

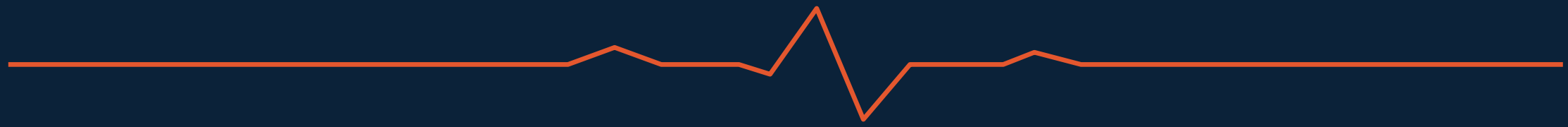
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github.com/Anurag-kushwaha2207/PulseForge_Pranveer-Singh-Institute-of-Technology



Thank you